

Hot water into a cold pipe

The physics of dead-leg warm-up, and how the deadleg model represents it

3 October 2026

When a hot tap is opened at the end of a pipe that has gone cold, the first water out is the cold water that was standing in the pipe. That much is simple displacement. But the hot water that follows has to heat the pipe wall as it goes, and it loses a little heat to the air. So the tap runs warm later, and more water is wasted, than displacement alone suggests. This note sets out the physics of that transient: the governing equations, the few dimensionless numbers that control it, the analytical limits, and why copper and plastic pipes behave so differently. The model built on it predicts the delivery times of 46 field tests to within a few seconds, with no fitted constants.

The question

Consider a pipe of length L (m) whose bore, the open channel the water flows through, has radius r_i (m). The pipe holds a volume of water $V = \pi r_i^2 L$ (m³). Let t (s) be the time since the tap was opened, and Q (m³/s) the volume flow rate through the pipe. If the hot water simply pushed the cold water out, like a piston, it would reach the tap after the *plug-flow time*

$$t_{\text{plug}} = \frac{V}{Q} = \frac{L}{u}, \quad u = \frac{Q}{\pi r_i^2}, \quad (1)$$

where u (m/s) is the mean velocity of the water.¹ The rest of this note explains why the real delivery time is longer than this, and by how much.

The setting is a domestic hot water branch. Flows are 1 to 6 l/min, through pipes of roughly 10 to 12 mm outside diameter, made of copper or plastic (PE-X, cross-linked polyethylene). Before the tap opens, the water in the pipe, the pipe wall and the surrounding air are all at one starting temperature, T_0 (°C), typically about 20 °C. From $t = 0$, water enters the pipe at the inlet temperature $T_{in}(t)$, for example 55 °C.

Four bodies, three exchanges

Let x (m) be the distance along the pipe from the inlet, and r (m) the radial distance from the pipe's axis. The pipe wall occupies $r_i < r < r_o$, where r_o is the outer radius. The model follows four bodies (Figure 1):

1. **the water**, flowing along the pipe as a plug. Its *bulk* temperature $T_w(x, t)$ is the flow-weighted average over the cross-section, which is the temperature you would measure if you collected and mixed the water passing x ;

¹ For 5 m of 10 mm copper (bore 8.8 mm), $V = 0.30$ l, so $t_{\text{plug}} = 6$ s at 3 l/min and 18 s at 1 l/min.

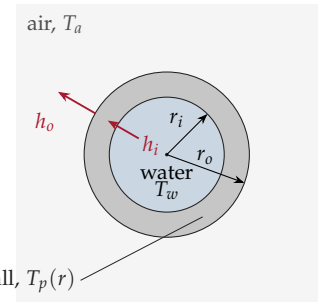


Figure 1: Cross-section of the pipe. Heat passes from the water to the wall across the water film (coefficient h_i), conducts through the wall, and leaves its outer surface for the air (coefficient h_o). r_i and r_o are the inner and outer radii.

2. **the pipe wall**, homogeneous and axially symmetric, with a temperature $T_p(r, x, t)$ that varies through the thickness and along the length (subscript p for *pipe*);
3. **the air**, a single well-mixed body at temperature $T_a(t)$ that warms as the pipe gives it heat;
4. optionally, **an environment**, such as the building fabric, at a fixed temperature T_{env} , to which the air body can lose heat.

They exchange heat in a chain. Heat passes from the water to the wall by forced convection, through the wall by conduction, and from the wall to the air by natural convection and radiation. A *heat transfer coefficient* ($\text{W}/\text{m}^2\text{K}$) is the heat flow per unit of surface area per degree of temperature difference. The coefficient between the water and the inner surface of the wall is h_i ; the coefficient between the outer surface and the air is h_o . The section on heat transfer coefficients, below, says how both are found.

Governing equations

Water. Let ρ_w (kg/m^3) and c_w ($\text{J}/\text{kg K}$) be the density and specific heat capacity of water, and $A_i = \pi r_i^2$ (m^2) the cross-sectional area of the bore. An energy balance on a thin slice of water, of length dx , carried along with the flow gives

$$\rho_w c_w A_i \left(\frac{\partial T_w}{\partial t} + u \frac{\partial T_w}{\partial x} \right) = h_i 2\pi r_i (T_p(r_i, x, t) - T_w). \quad (2)$$

The left side is the rate at which the slice's heat content changes, per metre of pipe, as seen by an observer moving with the water. The right side is the heat flowing into the slice from the wall, per metre: the coefficient h_i , times the wetted perimeter $2\pi r_i$, times the difference between the wall's inner-surface temperature $T_p(r_i, x, t)$ and the water temperature. The inlet condition is $T_w(0, t) = T_{in}(t)$, and the initial condition is $T_w(x, 0) = T_0$.²

Wall. Let ρ_p , c_p and k_p be the density, specific heat capacity and thermal conductivity of the pipe material.³ Heat conduction in an axially symmetric annulus obeys

$$\rho_p c_p \frac{\partial T_p}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(k_p r \frac{\partial T_p}{\partial r} \right) + k_p \frac{\partial^2 T_p}{\partial x^2}, \quad r_i < r < r_o. \quad (3)$$

The first term on the right is conduction through the thickness; the second is conduction along the pipe. The heat flux leaving a surface by conduction, $-k_p \partial T_p / \partial r$ (W/m^2), must equal the flux arriving there by convection, so the two faces obey

$$-k_p \frac{\partial T_p}{\partial r} \Big|_{r=r_i} = h_i (T_w - T_p(r_i)), \quad (4)$$

$$-k_p \frac{\partial T_p}{\partial r} \Big|_{r=r_o} = h_o (T_p(r_o) - T_a). \quad (5)$$

The wall starts at T_0 . Its ends, $x = 0$ and $x = L$, are treated as insulated.

² Two simplifications are made here. First, heat conduction along the water is neglected. The Péclet number $Pe = uL/\alpha_w$, the ratio of heat carried by the flow to heat conducted along it, is about 10^7 ; here $\alpha_w = k_w/(\rho_w c_w)$ is the thermal diffusivity of water and k_w ($\text{W}/\text{m K}$) its thermal conductivity. Second, $\rho_w c_w$ is held constant, at its value for the mean of T_0 and T_{in} , so that heat is carried along exactly. It varies by less than 1% over this range.

³ Here c_p means the specific heat of the *pipe* material, not the specific heat at constant pressure.

Air. Let ρ_a and c_a be the density and specific heat capacity of air, and V_a (m^3) the volume of the air body. Let UA_{env} (W/K) be the conductance between the air body and the environment: the heat flow per degree of temperature difference, which is zero by default. Because the air has a single temperature, its energy balance is an ordinary differential equation:

$$\rho_a c_a V_a \frac{dT_a}{dt} = \int_0^L h_o 2\pi r_o (T_p(r_o, x, t) - T_a) dx - UA_{env}(T_a - T_{env}). \quad (6)$$

The integral is the heat arriving from the whole outer surface of the pipe, which has perimeter $2\pi r_o$. The air starts at $T_a(0) = T_0$. Letting $V_a \rightarrow \infty$ holds the air at T_0 , which represents a large room.⁴

Conservation. Let $T_{out}(t) = T_w(L, t)$ be the temperature of the water leaving at the tap. Measure each body's heat content (J) from the starting temperature, so that each starts at zero:

$$\begin{aligned} E_{\text{water}} &= \int_0^L \rho_w c_w A_i (T_w - T_0) dx, \\ E_{\text{wall}} &= \int_0^L \int_{r_i}^{r_o} \rho_p c_p (T_p - T_0) 2\pi r dr dx, \\ E_{\text{air}} &= \rho_a c_a V_a (T_a - T_0). \end{aligned}$$

Adding equations (2)–(6) and integrating from 0 to t gives

$$\begin{aligned} \int_0^t \rho_w c_w Q (T_{in} - T_{out}) dt' &= E_{\text{water}} + E_{\text{wall}} + E_{\text{air}} \\ &+ \int_0^t UA_{env}(T_a - T_{env}) dt', \quad (7) \end{aligned}$$

where t' is the variable of integration. In words: the heat that entered and has not left at the tap is now in the water, the wall or the air, or has gone to the environment. The numerical solution satisfies this budget to rounding error.

What controls the delay

Writing equations (2)–(6) in dimensionless form shows that three dimensionless groups do almost all the work.

The heat capacity ratio. Let $A_p = \pi(r_o^2 - r_i^2)$ (m^2) be the cross-sectional area of the wall. The ratio

$$\sigma = \frac{\rho_p c_p A_p}{\rho_w c_w A_i} \quad (8)$$

is the heat one metre of wall absorbs per degree of warming, divided by the heat one metre of water gives up per degree. Suppose the wall took up heat instantly and nothing was lost to the air. For hot water to pass a point, the pipe there must first be raised to the hot temperature, which takes $(1 + \sigma)$ times the heat held in the water alone. The hot front, the position where the water changes from cold

⁴ c_a is the specific heat at constant pressure, 1006 J/kg K. That is appropriate for a void that is not airtight, where the warming air can expand. For a sealed void the specific heat at constant volume, about 0.72 times this, would apply instead.

to hot, would therefore travel at speed $u/(1 + \sigma)$. It would reach the tap at

$$t_{\text{front}} = (1 + \sigma) t_{\text{plug}}. \quad (9)$$

This is the most useful single result in this note. Copper's thin wall gives $\sigma \approx 0.2$; plastic's thick wall gives $\sigma \approx 0.5$ to 0.6 (Table 1).⁵

The number of transfer units. The group

$$NTU = \frac{h_i 2\pi r_i L}{\rho_w c_w Q} \quad (10)$$

is the heat-exchange capacity of the whole inner surface, in W/K, divided by the heat capacity flow of the water, also in W/K. It measures how completely the water gives its heat to the wall in one pass. If it were very large, the water and the inner surface of the wall would be at the same temperature everywhere. Over 5 m of these pipes it is between about 1 and 4. That is large enough for the wall to take most of the heat from the leading water. It is also small enough that some slightly warmed water arrives at the plug-flow time, ahead of the main rise. Above about 2 l/min, NTU hardly changes with flow, because h_i rises almost in proportion to u .

The wall Biot number and diffusion time. Let $\delta = r_o - r_i$ be the wall thickness and $\alpha_p = k_p/(\rho_p c_p)$ (m^2/s) the thermal diffusivity of the wall material. Then

$$Bi = \frac{h_i \delta}{k_p}, \quad t_{\text{wall}} = \frac{\delta^2}{\alpha_p}. \quad (11)$$

The Biot number Bi is the wall's resistance to conduction through its thickness, δ/k_p , divided by the water film's resistance, $1/h_i$. The diffusion time t_{wall} is roughly how long heat takes to soak through the thickness of the wall. This is where copper and plastic part company:

- copper: $Bi \sim 0.01$ and $t_{\text{wall}} \approx 4$ ms. The wall is at one temperature through its thickness almost as soon as the water touches it, so it behaves as a simple lumped heat store;
- PE-X: $Bi \sim 10$ to 60 and $t_{\text{wall}} \approx 12$ to 20 s. The wall is a poor conductor; while the hot water passes, only a thin inner skin warms.

	bore	σ	t_{wall}	NTU / Bi at		
	mm		s	1 l/min	3 l/min	6 l/min
Copper 10×0.6	8.8	0.24	0.004	2.1 / 0.002	3.4 / 0.009	3.2 / 0.02
Copper 12×0.6	10.8	0.19	0.004	1.3 / 0.001	2.8 / 0.006	2.7 / 0.01
PE-X 10×1.5	7.0	0.50	11.7	3.2 / 8	4.2 / 31	3.9 / 57
PE-X 12×2.0	8.0	0.59	20.8	2.5 / 7	3.7 / 33	3.4 / 60

Figure 2 shows the consequence. Copper absorbs its full share of heat at once, so the outlet stays cool for longer, and its rise is centred near the delayed front time of equation (9). PE-X takes only a skin of heat at first. Its *effective* σ , meaning the part of the wall that has had time to warm, is small early on and grows as heat penetrates. So the first warm water arrives close to the plug-flow time, but the last few degrees wait for the wall to soak through.

⁵ Per unit volume, copper stores only about 1.7 times as much heat per degree as PE-X ($\rho_p c_p$ of 3.4 against $2.0 \text{ MJ/m}^3\text{K}$). The difference in σ comes from the walls: a PE-X wall is 2.5 to 3.3 times as thick, and it surrounds a smaller bore.

Table 1: The controlling groups for a 5 m run. Pipes are named by outside diameter × wall thickness, in mm. Water properties are taken at 40°C , and NTU is for the whole 5 m. Copper wall: $k_p = 340 \text{ W/mK}$, $\rho_p c_p = 3.4 \text{ MJ/m}^3\text{K}$. PE-X: $k_p = 0.38 \text{ W/mK}$, $\rho_p c_p = 2.0 \text{ MJ/m}^3\text{K}$.

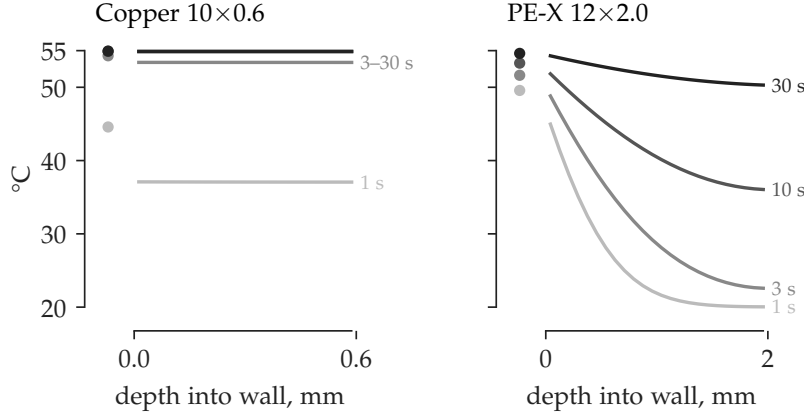


Figure 2: Temperature through the wall, half-way along a 5 m run at 3 l/min, at 1, 3, 10 and 30 s after the hot front passes. The dots at the left are the water. Copper is uniform through its thickness and catches up with the water in a few seconds. In PE-X a steep gradient persists: after 10 s the outer surface has warmed by only 16 K, while the water is 33 K above its starting temperature.

Heat transfer coefficients

Inside. The inside coefficient is found from three dimensionless numbers of the water: the Reynolds number $Re = uD_i/\nu_w$, the Prandtl number $Pr = \mu_w c_w/k_w$ and the Nusselt number $Nu = h_i D_i/k_w$. Here $D_i = 2r_i$ is the bore diameter, μ_w (Pa s) is the dynamic viscosity of water, and $\nu_w = \mu_w/\rho_w$ (m^2/s) its kinematic viscosity. Re compares inertia with viscosity and decides whether the flow is laminar or turbulent. Pr compares how fast momentum and heat diffuse. Nu is the heat transfer coefficient in dimensionless form. For fully turbulent flow, $Re \geq 10^4$, the model uses Gnielinski's correlation,

$$Nu_G = \frac{(f/8)(Re - 1000)Pr}{1 + 12.7\sqrt{f/8}(Pr^{2/3} - 1)}, \quad f = (0.790 \ln Re - 1.64)^{-2}, \quad (12)$$

where f is the Darcy friction factor of a smooth pipe. For laminar flow, $Re \leq 2300$, it uses $Nu = 3.66$, the value for fully developed laminar flow with a uniform wall temperature. Between the two, it interpolates as the VDI Heat Atlas recommends: $Nu = (1 - \gamma)3.66 + \gamma Nu_G(10^4)$. Here $Nu_G(10^4)$ is Gnielinski's value at $Re = 10^4$, and the weight is $\gamma = (Re - 2300)/(10^4 - 2300)$.⁶ This gives h_i of about 10^3 to $1.5 \times 10^4 \text{ W/m}^2\text{K}$.

Outside. Let $T_s = T_p(r_o)$ be the outer surface temperature and $D_o = 2r_o$ the outside diameter. The outside coefficient is the sum of a convective part and a radiative part, $h_o = h_{conv} + h_{rad}$. The convective part is natural convection from a horizontal cylinder, from the correlation of Churchill and Chu:

$$\frac{h_{conv} D_o}{k_a} = \left\{ 0.60 + \frac{0.387 Ra^{1/6}}{[1 + (0.559/Pr_a)^{9/16}]^{8/27}} \right\}^2, \quad (13)$$

$$Ra = \frac{g\beta |T_s - T_a| D_o^3}{\nu_a \alpha_a}. \quad (14)$$

Here Ra is the Rayleigh number, which measures the strength of buoyancy-driven flow. g is the acceleration due to gravity, and β

⁶ At 1–6 l/min in 7–11 mm bores, Re is about 3000–28000 at 40 °C, so the flow is transitional at the lowest flows and turbulent otherwise. The viscosity of water halves between 20 and 55 °C, so the arriving hot water is more turbulent than the cold water it displaces. The model therefore evaluates μ_w , k_w and Pr at the local water temperature.

(1/K) the expansion coefficient of air, taken as $1/T$ in kelvin. k_a , ν_a , α_a and Pr_a are the conductivity, kinematic viscosity, diffusivity and Prandtl number of air, at the mean of T_s and T_a . The radiative part treats the surroundings as a large enclosure at the air temperature:

$$h_{rad} = \varepsilon \sigma_{SB} (T_s^2 + T_a^2)(T_s + T_a), \quad (15)$$

with temperatures in kelvin. ε is the emissivity of the outer surface and $\sigma_{SB} = 5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$ the Stefan–Boltzmann constant. For a 50°C pipe in 20°C air, this gives $h_o \approx 9 \text{ W/m}^2\text{K}$ for tarnished copper ($\varepsilon = 0.1$) and $\approx 14 \text{ W/m}^2\text{K}$ for plastic ($\varepsilon = 0.9$).

The outside coefficient is 100 to 1000 times smaller than the inside one. While the tap runs, a warm pipe loses of the order of 10 W per metre to the air. The wall, meanwhile, absorbs heat at hundreds of watts per metre while the front passes. Over a draw of a minute or two, almost all of the heat leaving the water goes into the wall. External losses set the steady outlet temperature and how quickly the pipe cools between draws, but they barely change the delivery time. The model confirms this. Enclosing the pipe in a $1.5 \text{ m} \times 0.3 \text{ m}$ air body, instead of holding the air at a fixed temperature, changes no field-trial prediction by more than 0.34 s , even though the air warms by up to 7 K .

Exact solutions in limiting cases

Three limits have closed-form solutions. Each one checks the model and sharpens intuition.

No wall heat capacity. If $\rho_p c_p = 0$ and no heat is lost, the outlet steps from T_0 to T_{in} at t_{plug} , equation (1).

A thin, perfectly conducting wall with no losses. If k_p is so large that the wall is uniform through its thickness, and $h_o = 0$, the problem is the Anzelius–Schumann problem of regenerator theory. Define the dimensionless water temperature $\theta = (T_w - T_0)/(T_{in} - T_0)$, which runs from 0 (cold) to 1 (as hot as the supply). Let $P = 2\pi r_i$ be the wetted perimeter, and define the dimensionless distance and time

$$\xi = \frac{h_i P x}{\rho_w c_w Q}, \quad \eta = \frac{h_i P (t - x/u)}{\rho_p c_p A_p}.$$

ξ is the number of transfer units between the inlet and x . η is the time elapsed since the plug-flow front passed x , measured in units of the wall's thermal response time. For a step change in the inlet temperature at $t = 0$, and for $\eta \geq 0$,

$$\theta = e^{-\xi} \left[e^{-\eta} I_0(2\sqrt{\xi\eta}) + \int_0^\eta e^{-s} I_0(2\sqrt{\xi s}) ds \right]. \quad (16)$$

For $\eta < 0$, that is before the plug front arrives, $\theta = 0$. Here I_0 is the modified Bessel function of the first kind, of order zero, and s is a variable of integration. At the tap, $x = L$, so ξ equals the NTU . At

the moment the plug front arrives, $\eta = 0$, the bracket equals one. The outlet therefore jumps to $\theta = e^{-NTU}$: a small step of slightly warmed water. The rest of the rise is spread around t_{front} of equation (9).

Steady state with losses. After the wall has warmed fully, with the air at a fixed temperature, the outlet settles at

$$T_{\text{out}} = T_a + (T_{\text{in}} - T_a) \exp\left(-\frac{U'L}{\rho_w c_w Q}\right), \quad (17)$$

$$\frac{1}{U'} = \frac{1}{2\pi r_i h_i} + \frac{\ln(r_o/r_i)}{2\pi k_p} + \frac{1}{2\pi r_o h_o}. \quad (18)$$

U' (W/m K) is the overall conductance from water to air per metre of pipe: the inverse of the three resistances in series, namely the water film, the wall and the outside surface. Because h_o is small, the steady drop over 5 m is a fraction of a degree.

How the equations are solved

The pipe is divided into N_x cells of length $\Delta x = L/N_x$ (200 cells by default). Each cell has one water temperature and N_r concentric wall shells of equal thickness, each with its own temperature. There is one air temperature. Each time step, of length Δt , does three things:

1. **Radial heat exchange** between the water, the wall shells and the air, solved implicitly (backward Euler). In each cell the water and wall unknowns form a tridiagonal system, and its solution is linear in the new air temperature. That lets the air, which couples every cell, be solved exactly in one step, with no iteration;
2. **Axial conduction** along the wall, also implicit;
3. **Advection**, meaning transport by the flow: the time step is chosen as $\Delta t = \Delta x/u$, so the water moves exactly one cell per step. Transport is therefore exact, and the front is not artificially smeared.

The scheme conserves energy to rounding error. It converges to the Anzelius–Schumann solution at first order, meaning the error halves when the number of cells doubles. The largest error is 0.7% with 200 cells and 0.2% with 800 (Figure 3).

What the model predicts

Figure 4 shows both behaviours. Each copper trace stays at T_0 until t_{plug} and shows the small step of slightly warmed water. It then rises steadily over a few seconds, through $(1 + \sigma) t_{\text{plug}}$. The PE-X traces jump most of the way towards 40 °C at the plug-flow time, then creep towards the inlet temperature as heat soaks into the wall.

Three practical consequences follow.

- **For copper, the waste is a fixed volume.** About 1.4 to 1.5 pipe volumes run off before the tap reaches 50 °C, whatever the flow. The time therefore scales as $1/Q$.

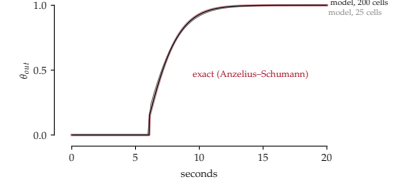


Figure 3: The numerical model (grey and black) against the exact Anzelius–Schumann solution (red), for a copper-like wall, 5 m at 3 l/min. θ_{out} is θ at the tap. Even with 25 cells the curves are almost indistinguishable.

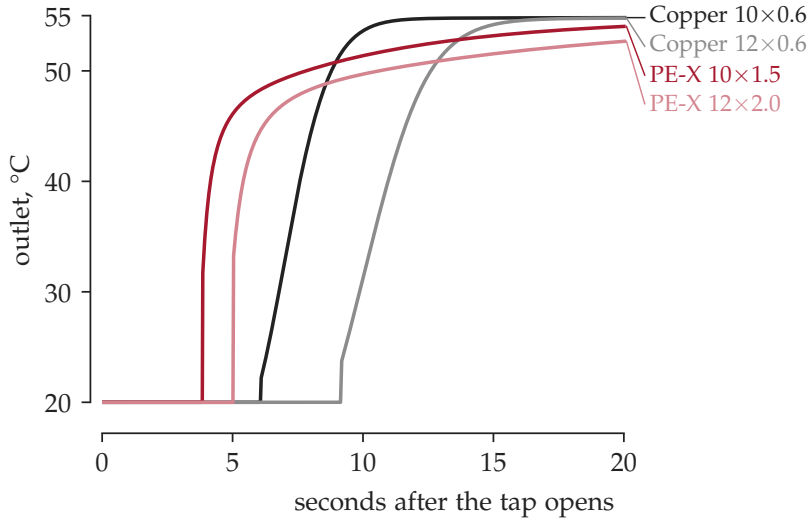


Figure 4: Outlet temperature after the tap opens: 5 m run, 3 l/min, inlet 55 °C, pipe and air initially 20 °C.

- **For plastic, the answer depends on what “hot” means.** The time to 50 °C can be twice the time to 40 °C. At 1 l/min, a PE-X pipe is as slow to 50 °C as a copper pipe of the same outside diameter, despite holding less water. At 6 l/min it is faster.
- **A delivery time means something only with its threshold and supply temperature.** A 45 °C target with a 50 °C supply falls in the slow tail for plastic pipe.

Checking against a field trial

Ridge and Jones (FairHeat, 2022) measured the time for the tap to reach 45 °C. They tested 5, 15 and 25 m runs of multilayer pipe (plastic with an aluminium layer) at 4, 6 and 9 l/min. Hot water came from a heat interface unit, the heat exchanger that serves a flat on a heat network, and the pipe started at about 19 °C. The probe was read every 5 s. The model was run for every test with no adjustment. The inlet temperature stepped from 19 to 50 °C at the time the unit alone had taken to deliver hot water, which the paper measured separately. The multilayer wall was represented as one homogenised material, and the air body was an enclosed 1.5 m × 0.3 m cross-section running the length of the pipe.

	model		paper’s regression	
	bias	RMS	bias	RMS
Manufacturer A (23 tests)	−0.1 s	3.3 s	+3.1 s	4.3 s
Manufacturer B, hold-out (23)	+1.1 s	7.0 s	+6.2 s	8.9 s

The physical model does better than the paper’s regression on both sets, including the set the regression was fitted to (Table 2, Figure 5). Its largest misses are on the slow, long, large-bore runs at 4 l/min, where it predicts hot water 10 to 28 s early. The likeliest explanation, which has not yet been tested, is that the heat unit sup-

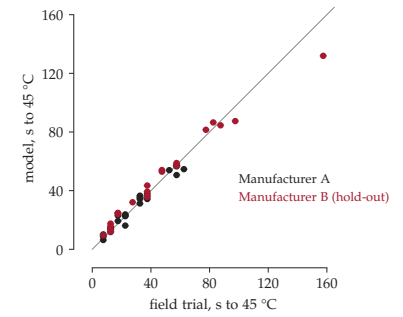


Figure 5: Model against measured time to 45 °C for all 46 field tests. The diagonal is perfect agreement. Each dot sits half-way through its 5 s reading window, and the short horizontal bar spans the window.

Table 2: Bias is the mean of predicted minus measured time. RMS is the root-mean-square error. Errors are taken against the middle of each 5 s reading window. The paper’s regression was fitted to Manufacturer A, so Manufacturer B is a hold-out set that neither method has seen.

plied water a little below 50 °C at that flow. Because 45 °C is close to the supply temperature, one degree makes a large difference there.

Limitations

- **Plug flow.** Below $Re \approx 2300$ the flow is laminar. Its velocity profile is parabolic, so water on the axis moves at twice the mean velocity. The first warm water then arrives at about half the plug-flow time, and the front is smeared. This matters only near or below 1 l/min, with cold water.
- **Steady correlations.** A coefficient h_i derived for steady conditions is applied to a strongly transient wall. The entrance region and the moving thermal front both raise the true coefficient slightly.
- **A homogeneous wall.** Multilayer pipe is represented by volume-averaged properties.
- **What is left out.** Fittings, clips, insulation and the thermal mass of the surrounding structure are not modelled. Radiation is taken to go to surroundings at the air temperature.

The code, tests and data behind this note are in the GitHub repository [gwilma/deadleg_deadtime](https://github.com/gwilma/deadleg_deadtime). The file `docs/derivation.md` gives the same derivation in brief, and `docs/physics/make_figures.py` redraws every figure.